Registration No.:_____ PNR No:: 11718ACAP50451

COURSE CODE : CAP538

COURSE NAME : ALGORITHM DESIGN AND ANALYSIS

Time Allowed: 03:00 hrs ***Max.Marks: 70***

Read the following instructions carefully before attempting the question paper.
1. This question paper is divided into two parts A and B.
2. Part A contains 10 questions of 2 marks each. All questions are compulsory.
3. Part B contains 5 questions of 10 marks each. In each question attempt either question (a) or (b). In case both (a) and (b) questions are attempted for any question then only the first attempted question will be evaluated.
4. Answer all questions in serial order.
5. Do not write or mark anything on the question paper except your registration no. on the designated space.

PART A

Q1 a)Illustrate the steps involved in analyzing algorithm using an example. [2 Marks]
b)What are the applications of data structures? [2 Marks]
c)Discuss the characteristics of greedy approach. [2 Marks]
d)What is greedy approach? what are the problem associated with greedy approach? [2 Marks]
e)Define the graph coloring problem? [2 Marks]
f)State the 0/1 KnapSack problem. [2 Marks]
g)What are the applications of Huffman coding? [2 Marks]
h)What do you mean by approximation algorithm? [2 Marks]
i)What is the purpose of lower bound theory? [2 Marks]
j)What is decision problem? Give an example. [2 Marks]

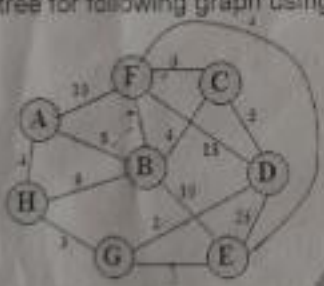
PART B

Q2 a)What is recurrence relation? How it is useful to analyze an algorithm. Illustrate with an example [10 Marks]

OR

b) (i)Analyze quick sort algorithm using recurrence equation. [5 Marks]
(ii)Write a note on Asymptotic notations [5 Marks]

Q3 a)What are the characteristics of minimum spanning tree? Generate the minimum spanning tree for following graph using Kruskal,s algorithm.



[10 Marks]

OR

b) (i)What is minimum spanning tree? Genrate the minimum spanning tree for following graph using Prims algorithm.



[5 Marks]
(ii)Write an algorithm to perform subtraction of two long integers. [5 Marks]

Q4 a)Consider the following 0/1 KnapSack problem instance:
N=4, (P1, P2, P3, P4) =(10, 10, 12, 18), (w1, w2, w3, w4)=(2, 4, 6, 9) and m= 15. Trace the working of least cost branch and bound. [10 Marks]

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OR

b) (i)Apply Knuth-Morris-Pratt technique on the following text and pattern:
Text: ababcdabbababababad
Pattern: abababa [5 Marks]

(ii)Generate the Huffman code for the text - "Grapes are sour". [5 Marks]

Q5 a)Discuss the problem of Bin Packing with suitable example. [10 Marks]

OR

b) (i)Match the following pattern with text using Boyer-Moore algorithm:
Text: ABCABCDABABCABAABDABCDABDE
Pattern: ABCDAB [5 Marks]

(ii)Draw a comparison tree for sorting any four numbers. Sort three elements A[1]=19, A[2]=16, A[3]=18 and A[4]=45 using comparison tree. [5 Marks]

Q6 a)Proof that satisfiability is in P if and if only if P= NP. [10 Marks]

OR

b) (i)Explain Cook's theorem in full detail. [5 Marks]

(ii)What is the relationship among P, NP, NP-complete and NP-hard problems? Explain with a neat and clean diagram. [5 Marks]

-- End of Question Paper --

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